

Priors, Biases, and Breakups: A Probabilistic Programming Approach to Attachment Dynamics

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Abstract

Romantic relationships depend heavily on how partners interpret each other’s behavior, form expectations, and update their beliefs over time. Attachment theory suggests that people with secure, anxious, and avoidant styles process the same signals in systematically different ways, often leading to unstable relationship dynamics. This project models these processes using a probabilistic framework in which partners receive noisy signals, apply style-specific interpretation biases, and adjust their beliefs about underlying compatibility. Simulations reveal how small differences in priors or learning rates can amplify over repeated interactions, producing outcomes such as premature breakups. These results highlight how cognitive biases, rather than “true” compatibility alone, can drive relationship trajectories and explain why some pairings are more fragile than others. More broadly, the work illustrates how probabilistic programming can translate psychological theories of attachment into models that capture patterns in human social behavior.

Keywords: attachment theory; Bayesian inference; probabilistic programming; relationship dynamics; cognitive biases

Related Works

Attachment theory describes how people form expectations and interpret signals in close relationships. In *Attached* (Levine & Heller,), Levine and Heller explain that secure individuals generally expect stability and interpret ambiguous behavior in a balanced way. Anxious individuals tend to overreact to negative cues and treat neutral signals as possible threats. Avoidant individuals under react to positive cues and often view closeness as uncomfortable. These patterns appear in my model as priors, interpretation biases, and asymmetric learning rates. Rather than describing attachment styles qualitatively, I quantify how they influence relationship dynamics by formalizing each style as a distinct belief-updating process over a shared compatibility variable.

To understand Bayesian approaches to social learning, I looked at Lamba et al. (Lamba, Houlihan, & Saxe,), a study on how people adjust their behavior when repeatedly interacting with partners who create unequal outcomes. They show that Bayesian updating can explain how people learn about a partner’s strategy over time and how psychological traits like social anxiety shape compliance with unfair situations. This work demonstrated that belief updating can capture systematic patterns in social interactions and helped motivate the learning rules in my simulation.

A related area of computational social reasoning examines how people infer hidden traits during decision making.

Xiang, Velez, and Gershman (Xiang, Velez, & Gershman,) model how individuals estimate whether potential teammates will be reliable or cooperative by updating beliefs about unobserved traits. Their approach informed how I represent compatibility in my model, treating it as a latent variable that must be inferred from signals rather than directly observed.

Although these studies shaped the main components of my model, my approach differs in several ways. Prior work typically uses one-directional inference, where a single agent updates beliefs about many possible partners or responds to fixed strategies. In contrast, my simulation involves two agents who simultaneously update their beliefs about each other, resulting in feedback loops. By integrating attachment theory with two-sided Bayesian learning, my project examines how mutual inference can lead to relationship stability or breakdown depending on the combination of attachment styles.

Methods

Overview

I built a two-agent probabilistic model in WebPPL in which two partners interact repeatedly over a fixed horizon $T = 20$. Each partner has a fixed attachment style (secure, anxious, or avoidant), and both agents update their beliefs about a shared latent compatibility variable C . Each round produces (a) a noisy signal generated from C , (b) an attachment-style-specific interpretation of that signal, (c) belief updates for each agent, and (d) stay/leave decisions. The model follows a generative process of the form:

$$C \rightarrow \text{signal} \rightarrow \text{interpreted signal} \rightarrow \text{belief update} \rightarrow \text{decision}.$$

I analyze behavior by running forward simulations and computing expectations over many relationship trajectories.

Attachment Style Parameters

Each attachment style is defined by a set of parameters that determine how the agent interprets signals and how strongly they adjust their beliefs over time. For each style s , I specify:

Prior beliefs (μ_{prior} and σ_{prior}) μ_{prior} is the agent’s starting belief about how good the relationship is, with a higher value meaning the agent begins more optimistic. σ_{prior} determines how certain the agent feels about that belief.

Interpretation biases (biasPos, biasNeutral, biasNeg)

Each raw signal takes a value in $\{-1, 0, +1\}$, and each style adds a bias before learning from it. Anxious agents shift neutral signals slightly negative. Avoidant agents shift strong positive signals downward. Secure agents add no bias.

Learning rates (λ_{pos} and λ_{neg}) A learning rate controls how much the belief moves after each interaction. Avoidant agents have lower λ_{pos} , meaning they barely update from positive signals. Anxious agents have a much larger λ_{neg} than λ_{pos} , meaning negative experiences affect them more strongly than positive ones.

Decision temperature (τ_s) This parameter controls how noisy or deterministic stay/leave decisions are. A low τ_s leads to decisions tightly aligned with the current belief, while a high τ_s introduces more randomness.

Leave threshold (θ) This is the belief level below which the agent decides to leave the relationship. Secure agents tolerate some negativity; anxious agents tolerate a great deal; avoidant agents leave as soon as the relationship stops looking clearly positive.

Table 1: Attachment Style Parameters. Each attachment style has a distinct set of priors, interpretation biases, and learning rates.

Parameter	Secure	Anxious	Avoidant
μ_{prior}	0.5	0.2	-0.2
σ_{prior}	0.75	1.2	0.9
$bias_{pos}$	0.0	-0.1	-0.4
$bias_{neutral}$	0.0	-0.3	0.1
$bias_{neg}$	0.0	-0.2	0.0
λ_{pos}	0.4	0.3	0.2
λ_{neg}	0.4	0.8	0.5
θ	-0.5	-0.8	0.0
τ_s	1.0	0.7	1.3

Model Implementation Structure

I organized the model across four WebPPL files: attachment.wppl, compatibility.wppl, dynamics.wppl, and main.wppl. Each file handles a different part of the generative process, making it easier to experiment with specific components.

attachment.wppl defines the attachment styles and all parameters described above. This file isolates psychological assumptions so that the remaining code interacts only with style parameters. It contains:

- `makeAttachmentStyle()`: builds a style object from parameter values;
- `secureStyle`, `anxiousStyle`, `avoidantStyle`: the three style definitions;

- `getAttachmentStyle(name)`: retrieves a style object by name.

compatibility.wppl defines how compatibility produces signals, and how those signals are interpreted, which is the environment side of the model. Structuring the environment in its own file helps preserve a clean separation between what the world generates and how agents interpret it. This follows Marr’s idea that the statistical structure of the world should be modeled independently from the cognitive processes acting on it. It contains:

- `sampleCompatibility()`: draws the true compatibility value C from a normal distribution. This is the hidden variable both agents are trying to estimate.
- `baseSignalDistribution(C)`: defines how likely each raw signal is given the true compatibility C . A higher C is more likely to see positive (+1) signals while a lower C is more likely to see negative (-1) signals. I then used a sigmoid transformation to make the relationship between C and signal likelihood smooth rather than binary.
- `sampleRawSignal(C)`, `interpretSignal(style, rawSignal)`, and `sampleRawSignalWithShock(C)`: these functions combine the environment’s raw signals with style-specific interpretation biases. This structure separates the generation of evidence (based on C) from the psychological interpretation of that evidence (based on attachment style). This maps directly onto Marr’s levels of analysis: one level describes the statistical structure of the world and the other describes how agents encode that structure.
- `sampleShock()`: adds rare events that override normal signal generation. Shocks allowed me to test how agents react to sudden, surprising information rather than only gradual evidence. In the real world, this translates to, for example, your significant other forgetting your anniversary.

dynamics.wppl defines how beliefs change over time and how decisions are made. It contains:

- `updateBelief(mu, style, s)`: implements the learning rate rules. If the signal is positive, add $\lambda_{pos} \cdot s$. If the signal is negative, add $\lambda_{neg} \cdot s$. If the signal is neutral, no update. This rule encodes the idea that an anxious person reacts quickly to negative signals and an avoidant person absorbs positive signals slowly.
- `shouldLeaveDeterministic(style, mu)` and `shouldLeaveStochastic(style, mu)`: the two decision modes. Both compare the agent’s current belief μ to their threshold θ . The stochastic version uses a softmax so decisions can have randomness. Most experiments use the deterministic version to focus on the effect of beliefs themselves.

- `simulateRelationship(styleA, styleB, options)`: a recursive simulation of an entire relationship without loops. It (a) generates signals for each agent, (b) applies interpretation biases, (c) updates each agent's belief, (d) checks stay/leave conditions (e) records the full history (signals, beliefs, and decision timing). This structure mirrors a typical Bayesian generative model: each step produces a new belief state that feeds into the next round.

main.wpp1 runs simulations and computes statistics. After separating psychological assumptions, environmental structure, belief dynamics, the model was easy to modify. It includes:

- `runDemoTrajectory()`: prints belief histories and decisions.
- `summarizePairing()`: runs hundreds of simulations per style pairing and computes breakup rates, false breakup probability, belief accuracy (MSE), mean leave timing, and average experienced signals. The file concludes by running experiments for all six pairings.

Code Availability

All code used for simulations and analyses is available at: [GitHub Repository](#)

Results

Across 400 simulations per attachment-style pairing, clear behavioral patterns emerged. Although the model uses simple learning rules, the six dyads behaved noticeably differently once attachment-specific priors and noisy signals were introduced. Across all analyses, a consistent theme emerged: relationship outcomes were driven far more by the styles' learning dynamics than by the latent compatibility value itself.

Breakup Rates Across Attachment Pairings

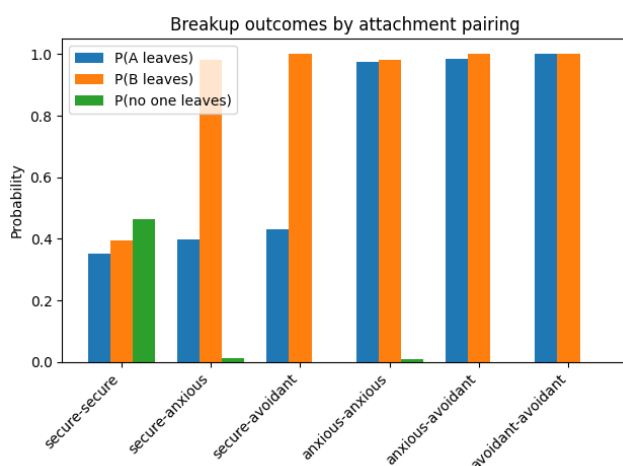


Figure 1: Probability that partner A leaves, partner B leaves, or both stay together across attachment-style pairings.

Secure-secure was the only pairing that showed a meaningful probability of staying together (about 46%). All other pairs involving at least one insecure partner broke up almost every time. Pairings with avoidant partners reached nearly 100% breakup probability within the 20-round window. This reflects how avoidant priors (high thresholds, pessimistic expectations) combine with noisy interactions to quickly trigger an exit, even when the underlying compatibility is neutral or positive.

Interestingly, secure-anxious and secure-avoidant pairs did not look substantially safer than anxious-anxious or anxious-avoidant pairs. The model suggests that a single insecure learning pattern can destabilize the entire relationship.

When Partners Choose to Leave

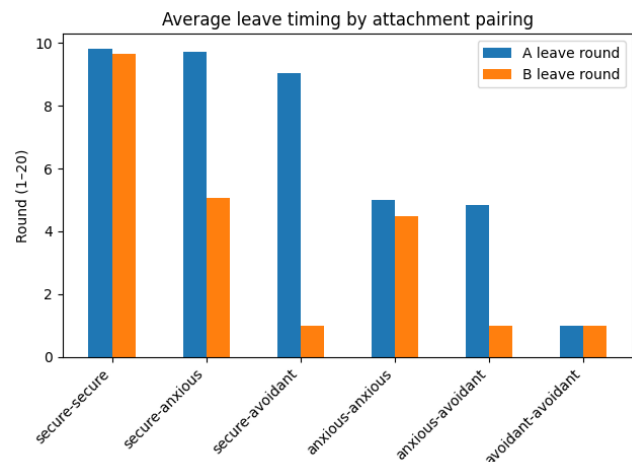


Figure 2: Average round in which each partner chose to leave (conditional on leaving).

Secure partners tended to leave late in the interaction (rounds 9–10), reflecting their patient and optimistic updating. Avoidant partners often left immediately, usually within the first round. Anxious partners fell in the middle, leaving after a few negative signals (around rounds 4–5). These timing differences show how small variations in interpretation can produce very different behavioral trajectories.

Misaligned Outcomes: False Breakups and Staying in Bad Relationships

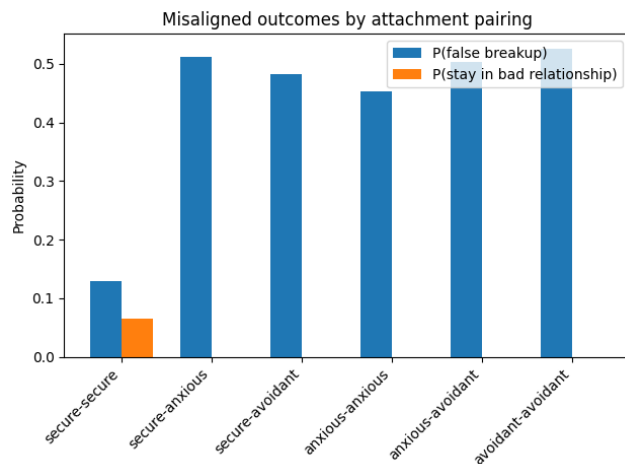


Figure 3: Probability of false breakup (ending a relationship despite positive compatibility) and probability of staying in a bad relationship.

Mixed-style pairs showed extremely high false-breakup rates. Secure-anxious, secure-avoidant, and anxious-avoidant pairs all had false-breakup probabilities near 0.50. In other words, half the time, partners abandoned relationships that were actually good. Secure-secure couples were much more resilient, with only 13% false breakups, suggesting that matching learning rules—not compatibility itself—support stability.

Belief Accuracy: Learning the True Compatibility

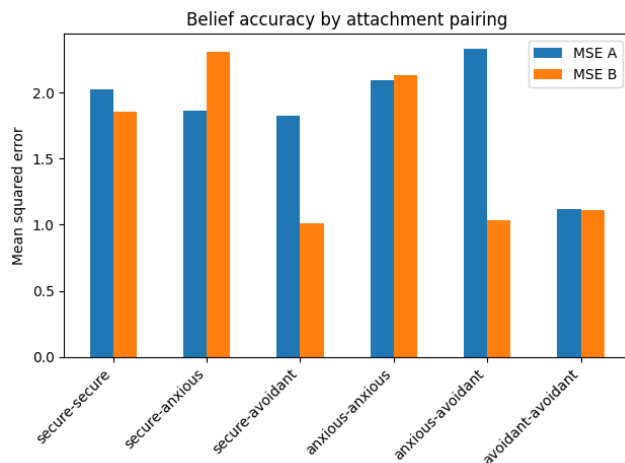


Figure 4: Mean squared error between each partner’s final belief and the true underlying compatibility.

Accuracy varied dramatically across attachment styles. Anxious partners showed the highest error, often overreacting to

negative noise. Avoidant partners were the most accurate overall, updating conservatively and smoothing over short-term fluctuations. Secure partners fell in the middle. These results support the idea that attachment styles function as perceptual filters that influence how much evidence agents absorb from each interaction.

Average Experienced Signals

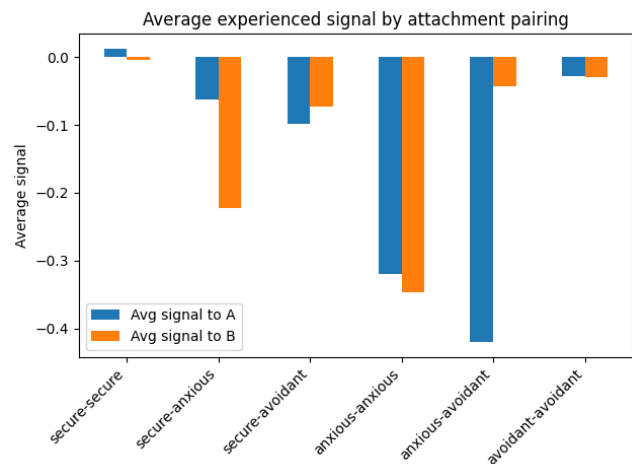


Figure 5: Average valence of signals experienced by each partner across the interaction.

The model shows that insecure pairings generate different emotional climates. Secure-secure interactions were neutral on average, suggesting balanced reinforcement. Anxious-anxious and anxious-avoidant pairs produced strongly negative signal streams, even when latent compatibility was not negative, reflecting a feedback loop where misinterpretation amplifies tension. Avoidant-avoidant dyads received nearly neutral signals but still dissolved quickly, indicating that threshold-driven behavior can override the content of the interaction.

Summary

Taken together, these results show that attachment-specific learning rules can strongly shape relationship outcomes even when partners are equally compatible. Secure-secure pairs thrive because they interpret signals similarly and update in stabilizing ways. Insecure combinations produce unnecessary breakups, distorted beliefs, and emotional spirals that diverge sharply across pairings.

Discussion

This project started from a simple question: if two people with different attachment styles are equally “good” for each other on paper, can their learning rules alone push the relationship toward success or failure? The simulations suggest the answer is yes. Across pairings, outcomes were shaped much more by how partners interpreted and updated signals

than by the underlying compatibility value. Secure–secure pairings stayed together not because they were more compatible, but because they shared moderate priors, symmetric learning rates, and similar thresholds. Their beliefs moved in step, which stabilized the relationship.

In contrast, most mixed-style pairs failed for reasons that had little to do with compatibility. Anxious agents reacted strongly to negative evidence, while avoidant agents treated positive signals as intrusive and left quickly once their beliefs slipped below zero. When these styles interacted, their parameters pushed them in opposite directions. The anxious partner became increasingly pessimistic, while the avoidant partner saw the anxious behavior as confirmation that something was wrong. The model shows how this kind of feedback loop can make “false breakups” common: relationships end even when the latent compatibility would have supported them.

One surprising result was that avoidant partners were the most accurate in estimating the true compatibility value, yet they were also the quickest to leave. Their conservative learning rates smoothed out noise, so their final beliefs were often close to the truth. However, their leave threshold was set so high that they walked away from many relationships that were objectively fine. This highlights a broader point discussed in class: a good internal model of the world does not guarantee good decisions if the decision rule is misaligned with the person’s goals or values.

Another notable pattern was that the model almost never produced cases of “staying in a bad relationship.” Agents were much more likely to exit too early than to remain in a truly negative pairing. This is partly a consequence of the parameter choices: anxious agents eventually react to repeated negative evidence, and avoidant agents leave as soon as the relationship stops looking clearly positive. In real life, people sometimes stay in harmful relationships because of external constraints or additional motives (financial dependence, fear, children) that are not captured here. The model is closer to a “clean lab” version of relationship cognition, where only beliefs about compatibility matter.

These strengths come with clear limitations. First, all parameters were hand-tuned based on qualitative descriptions from Attached (Levine & Heller,) and general intuitions, not fit to human data. A more rigorous approach would estimate priors, learning rates, and thresholds from studies of couples with known attachment styles. Second, compatibility is reduced to a single scalar. Real relationships involve multiple latent dimensions (trust, conflict style, shared goals), which could be modeled with a vector of hidden variables rather than one C . Third, partners in the model do not choose actions strategically. They only receive signals and decide whether to stay or leave. In reality, people also choose what messages to send, which could be modeled with an explicit planning or MDP layer.

The way shocks are implemented is also simplistic. Sudden positive or negative events are just rare ± 1 overrides, with

no memory beyond their effect on beliefs. In real relationships, certain shocks (for example, cheating or a major act of support) may permanently change how future signals are interpreted. A more realistic model could let shocks update not only momentary beliefs about compatibility, but also the attachment-style parameters themselves. That would connect this work to models of developmental change and “learning to learn.”

Despite these simplifications, the project illustrates how probabilistic programming can turn verbal theories of attachment into explicit computational claims. By writing down priors, biases, learning rates, and thresholds, I had to commit to specific mechanisms rather than broad labels like “clingy” or “avoidant.” This made it possible to generate quantitative predictions, such as the relative stability of secure–secure pairings or the high false-breakup rate for mixed pairs. The model also shows how two-sided Bayesian learning naturally creates feedback loops, which is harder to see in one-sided partner-evaluation models.

Future work could go in several directions. On the cognitive side, the next step would be to compare the model’s predictions to real data, such as breakup rates and satisfaction scores for different attachment-style pairings. On the modeling side, it would be interesting to let agents adapt their behavior, not just their beliefs, and to explore hierarchical Bayesian versions where priors over compatibility are learned across multiple relationships. Even with its current simplicity, the model supports a useful takeaway: relationship outcomes may depend less on who people are “meant” to be with and more on how their internal learning systems respond to the noise and ambiguity of everyday interaction.

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